# Contents


**1 Prime factorization** **1**

**2 Sums of polygon angles** **2**

**3 Area of a circle** **3**

**4 Polynomial theorems** **4**
- 4.1 The factor theorem . . . 4
- 4.2 The remainder theorem . . . 4

**5 Differentiation** **4**
- 5.1 Continuous functions . . . 4
- 5.2 The chain rule . . . 5
- 5.3 e . . . 5
- 5.4 The power rule . . . 6

**6 Integration** **7**
- 6.1 Riemann integration . . . 7
- 6.2 Fundamental theorem of calculus . . . 8
- 6.3 Integration by substitution . . . 9

**7 Differential equations** **9**

**8 Geometric series** **10**

**9 Surface areas and volume** **10**

**10 Introduction to complex numbers** **11**

**11 Trigonometry** **12**
- 11.1 Addition formulae . . . 12
- 11.2 Limits . . . 12

**12 Dot products** **13**

**13 The generalized binomial theorem** **14**
- 13.1 Introduction to taylor series . . . 14
- 13.2 The proof . . . 15
- 13.3 Other taylor series . . . 16

**14 Partial fractions** **16**

**15 Inverses of differentiable functions** **19**

**16 Cobweb diagrams and newton raphson** **20**

**17 Exponentials and logarithms** **20**

**18 (Appendix A) Motivation for the exponentials and logarithms video** **20**

**19 (Appendix B) Technical details from section 13** **20**

**20 (Appendix C) The Lebesgue integral** **23**


# 1 Prime factorization

**Theorem.** Every positive integer can be factored into primes, and furthermore, this can be done in only one way.

*Proof.* Clearly for every number a factorization into primes exists – I can take any number and make a factorization by repeatedly factoring its factors until I can't anymore, which will be when the factors are prime. Now I will prove that this can be done in a unique way. I use the fact that clearly 1 and 2 have a unique factorization (1 factors as no primes and 2 factors as just 2) and I show that all numbers have a unique factorization by showing that if all numbers $2, 3, 4, \ldots, n$ have a unique factorization so does $n+1$, as then 2 having one implies 3 does, 2 and 3 having one implies 4 does, etc. This type of logic is known as induction.

Suppose that each of the numbers $2, 3, 4, ..., n$ has a unique prime factorization. We must show that so does $n+1$. So, suppose for a contradiction that this is not true and that

$$n + 1 = p_1 p_2 ... p_s = q_1 q_2 ... q_t$$

where each of the p's and q's are prime. We will reorder each side such that the primes are in ascending order of size.

First, suppose that we are in the case where $p_1 = q_1$.

In this case, we can cancel it from both sides to get $p_2 p_3 ... p_s = q_2 q_3 ... q_t$. Since these are both less than or equal to n, it follows that the primes must be the same, since we are assuming that each of the numbers $2, 3, 4, ..., n$ has a unique prime factorization.

Therefore, we will assume that $p_1 < q_1$. If we can prove it, we will be done, because the $q_1 < p_1$ case is the same argument just with the roles of p and q swapped.

In this case, consider the number

$$x := p_1(p_2 p_3 ... p_s - q_2 q_3 ... q_t)$$

Clearly this is less than or equal to n since it is less than $p_1 p_2 ... p_s$. It is also positive since $p_1 < q_1$ implies that $\frac{n+1}{p_1} > \frac{n+1}{q_1}$ which in turn implies that $p_2 p_3 ... p_s > q_2 q_3 ... q_t$.

Now

$$x = p_1 p_2 p_3 ... p_s - p_1 q_2 q_3 ... q_t = q_1 q_2 ... q_t - p_1 q_2 q_3 ... q_t = (q_1 - p_1) q_2 q_3 ... q_t$$

Since $x \leq n, x$ has unique prime factorization. Therefore, since x is divisible by $p_1$ from how we defined it, and by assumption

$$q_t \geq q_{t-1} ... \geq q_2 \geq q_1 > p_1$$

the only place that the $p_1$ can appear in the factorization in

$$x = (q_1 - p_1) q_2 q_3 ... q_t$$

is in the $(q_1 - p_1)$ part (since we have uniqueness as $x \leq n$). However, this implies that $q_1$ is divisible by $p_1$ while not being equal to it, contradicting primality.

□

# 2 Sums of polygon angles

**Theorem.** Suppose a polygon has n sides, then its interior angles add up to $180(n-2)$.

*Proof.* We use the same principle of induction that we did for the prime factorization proof. We want to show that any n sided polygon can be triangulated into $n-2$ triangles like in figure 1, where each vertex of each triangle is a vertex of the original polygon.

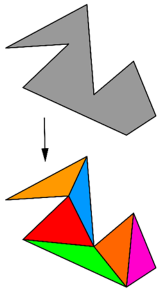
Figure 1

If we can prove this triangulation claim, we see that the sum of the interior angles becomes $180(n-2)$ as the sum of the interior angles of the original polygon must be equal to the sum of the interior angles of the $n-2$ triangles, which clearly is $180(n-2)$ by the convex case we proved in level 2 as triangles are indeed convex. Now it remains to show that a triangulation exists. The approach will be to show that if a triangulation exists into a number of triangles equal to the number of sides minus 2 for any polygon with 3, 4, 5, ... n-1 sides, then it works for n sides as well, as then we can say that well clearly it works for triangles so it must work for four sided shapes, but it works for triangles and four sided shapes so it must work for five sided shapes, and so on until we have it for all numbers of sides.

Now, suppose all polygons with fewer than n sides can be triangulated into the number of sides minus two triangles, and that we have a polygon with n sides. consider the leftmost vertex. Clearly, the angle at that vertex is less than 180 degrees, since otherwise it wouldn't be the leftmost vertex! (Just think about it hard enough to see this). Then consider the two neighbouring vertices and try to draw a line between them. Possibly this line does not intersect part of the polygon, in which case we form a triangle and have successfully split it into a triangle and another polygon with $n-1$ sides which by our assumption about any polygon with fewer than $n$ sides can be triangulated into $n-3$ triangles, so our original polygon can be triangulated into $n-2$ triangles, so done (Figure 2 shows this case):

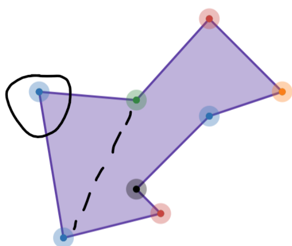

Figure 2

Otherwise, we connect the two neighbouring vertices to the leftmost one and part of the polygon is inside the triangle formed. In which case, we pick the leftmost vertex inside this triangle (meaning no part of the polygon is in any point in the triangle to the left) and connect it to the leftmost vertex, noting that now this diagonal cannot be obstructed, so we have split the polygon. Suppose it has been split into one with x sides, then the other one has $n+2-x$ sides (As the total number of sides of the two becomes $n-2$ as the diagonal which we used to cut used to contribute 0 sides and now contributes 2). By our assumption about polygons with fewer than n sides, we have that one part can be triangulated into $x-2$ triangles and the other part can be triangulated into $n-x$ triangles, so the original polygon can be triangulated into $n-2$ triangles (Figure 3 shows this case):

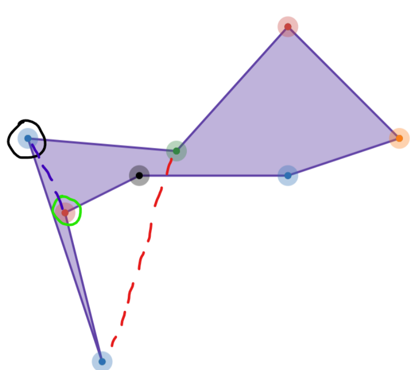

Figure 3

□

## 3 Area of a circle

**Theorem.** Suppose a circle has radius r, then its area is $\pi r^2$.

*Proof.* In figure 4, you can see a circle cut into many slices and then below shows the slices rearranged to form a rectangle-looking shape of height $r$ and width $\pi r$.

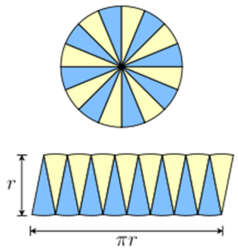

Figure 4

From this diagram it is clear that as we make the slices smaller the area of the figure on the bottom approaches a rectangle which will clearly have an area of $\pi r^2$.

□

# 4 Polynomial theorems

## 4.1 The factor theorem

**Theorem.** (The factor theorem) Given a polynomial $f(x), x-a$ divides $f(x)$ if and only if a is a root of $f(x)$, ie $f(a)=0$

*Proof.* First, notice the wording "if and only if". This means that the statements are equivalent, which means one implies the other. Therefore, it is sufficient to show that $x-a$ dividing $f(x)$ implies $f(a)=0$ and that $f(a)=0$ implies that $x-a$ divides $f(x)$.

First, suppose $x-a$ divides $f(x)$. This means that there exists a polynomial $g(x)$ such that $f(x)=(x-a)g(x)$. Now evaluate $f(a)$ by substituting a in place of x. This gives $f(a)=(a-a)g(a)$. Since $a-a=0$, $f(a)=0*g(a)=0$.

Now suppose that $f(a)=0$. We know that if we went through the long division process, we would get that $f(x)=(x-a)g(x)+R$ for some polynomial $g(a)$ and some remainder R. Since this is true for all values of x, it is true when $x=a$, so $f(a)=(a-a)g(a)+R$, but we know that $(a-a)g(a)=0$ since $a-a=0$ and that $f(a)=0$ by our assumption, so $0=0+R$, therefore $R=0$. This means that $f(x)=(x-a)g(x)$ so $x-a$ divides $f(x)$.

□

## 4.2 The remainder theorem

**Theorem.** (The remainder theorem) Given a polynomial $f(x)$, the remainder when you divide by $x-a$ is equal to $f(a)$. Furthermore, if you divide $f(x)$ by $bx-a$ then the remainder is$f\left(\frac{a}{b}\right)$.

*Proof.* First, note that $f(x)=(x-a)g(x)+R$ is true for all values of x, so substituting $x=a$ gives $f(a)=(a-a)g(a)+R$. Since $(a-a)g(a)=0$ since $a-a=0, f(a)=R$.

If we divide $f(x)$ by $bx-a$ we will get $f(x)=(bx-a)g(x)+R$ for some polynomial g. Let $x=\frac{a}{b}$ since this is true for all x, then we get $f\left(\frac{a}{b}\right)=\left(b\left(\frac{a}{b}\right)-a\right)g\left(\frac{a}{b}\right)+R=(a-a)g\left(\frac{a}{b}\right)+R$ (since the b's cancel in the $b(a/b)$ term) $=R$, since again, $(a-a)$ times anything is always 0.

□

# 5 Differentiation

## 5.1 Continuous functions

We want to define what it means for a function to be continuous.

Specifically, we want to capture the "you can draw it without taking your pen off the paper" intuition with a proper definition. The idea is that we say the function is continuous if the function is arbitrarily close to its value at some input x, provided the input is sufficiently close to x.

**Proposition.** If a sequence converges, ie $x_n \to x$, then $f(x_n) \to f(x)$ if $f$ is continuous at x.

*Proof.* $x_n$ can be made arbitrarily close to $x$ by the fact that the limit exists, and by continuity that means we can make $x_n$ close enough to $x$ that by making n large enough we can make $f(x_n)$ as close as we want to $f(x)$ so $f(x_n) \to f(x)$.

□

We will translate "arbitrarily close" and "sufficiently close" into mathematical language.

**Definition.** A function is said to be continuous at a point $x$ if for every $\varepsilon > 0$ (ie, a threshold that can get as small as you like), there exists a $\delta > 0$ (ie, your "sufficiently close") such that if $|x - y| < \delta$ then $|f(x) - f(y)| < \varepsilon$.

## 5.2 The chain rule

$$\frac{d}{dx} g(f(x)) = g'(f(x))f'(x)$$

*Proof.* I will start by giving a false proof.

$$\frac{d}{dx} g(f(x)) = \lim_{h\to 0} \frac{g(f(x+h)) - g(f(x))}{h} = \lim_{h\to 0} \frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} * \frac{f(x+h) - f(x)}{h} = g'(f(x))f'(x)$$

The reason this proof is false is because of the worry that $f(x+h) - f(x)$ could be 0. Here is the proper proof.

Define a new function $d(y) := \begin{cases} \frac{g(y)-g(f(x))}{y-f(x)} & y \neq f(x) \\ g'(f(x)) & y = f(x) \end{cases}$

By the definition of the derivative, the function d is continuous at $y = f(x)$ because $\frac{g(y)-g(f(x))}{y-f(x)} \to g'(f(x))$ as $y \to f(x)$.

If $g(f(x+h)) \neq g(f(x))$ then by the above false proof, we can actually say that

$$\frac{g(f(x+h)) - g(f(x))}{h} = d(f(x+h)) \frac{f(x+h) - f(x)}{h} \tag{1}$$

if $g(f(x+h)) = g(f(x))$ then equation (1) becomes $0 = 0$ and is therefore still true. Therefore, we can say that

$$\frac{d}{dx} g(f(x)) = \lim_{h\to 0} d(f(x+h)) \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} d(f(x+h)) \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = g'(f(x))f'(x)$$

□

## 5.3 e

**Lemma.** You can actually differentiate $c^x$ if $c > 1$.

*Proof.* Skip this if plotting it and looking is convincing enough to you. Otherwise,

The derivative should be

$$\lim_{h\to 0} \frac{c^{x+h} - c^x}{h} = c^x \lim_{h\to 0} \frac{c^h - 1}{h}$$

It therefore suffices to show that $\lim_{h\to 0} \frac{c^h-1}{h}$exists, ie that there is some number that this approaches as h tends to 0.

If $h > 0$ then the numerator and the denominator are both positive as $c^h > c^0 = 1$ since $c > 1$. We will show that $\frac{c^h-1}{h}$is an increasing function for positive rational numbers h and then since it is bounded below by 0, it must have some "highest lower bound" that it approaches as h goes to 0. This justifies existence of the limit since rational numbers are everywhere (ie, you can find them on any interval of the number line) and the existence of a "highest lower bound" is usually part of the definition of the real numbers and not something we need to justify, it's intuitive why this is the case. Note that since the rational numbers are everywhere and $\frac{c^h-1}{h}$ is clearly continuous at each $h > 0$, it indeed suffices to show that $\frac{c^h-1}{h}$ increases as h increases assuming h is rational.

Let a and b be positive rational numbers and let k be their common denominator so that $a = \frac{n}{k}, b = \frac{m}{k}$, and n and m are positive integers. Define $x := \frac{1}{k}$. Then we have to prove that $a > b$ implies $\frac{c^a-1}{a} > \frac{c^b-1}{b}$, or equivalently that $a > b$ (so $n > m$) implies $\frac{c^{nx}-1}{nx} > \frac{c^{mx}-1}{mx}$. We can use a factorization trick to write

$$\frac{c^{nx}-1}{nx} = \frac{c^x-1}{x} * \frac{1 + c^x + c^{2x} + ... + c^{(n-1)x}}{n}$$
$$\frac{c^{mx}-1}{mx} = \frac{c^x-1}{x} * \frac{1 + c^x + c^{2x} + ... + c^{(m-1)x}}{m}$$

We see that indeed, the first expression is larger since to compare them we can cancel the common non-zero factor $\frac{c^x-1}{x}$. But now we are comparing the means of two sets of terms, as that is exactly what the remaining fractions are, but $n > m$ by hypothesis so the one with n terms so the fraction with n terms simply has the same terms as the one with m terms and more terms which are larger, so the terms have a greater mean.

Note that we only addressed the $h > 0$ case. If $h < 0$, then set $k := -h$, then we are looking for the limit of $\frac{c^h-1}{h} = \frac{c^{-k}-1}{-k} = c^{-k}\frac{c^k-1}{k}$ as k goes to 0 where $k \to 0$. But since $c^{-k} \to 1$, the limit here is the same as the limit we found before.

□

**Theorem.** e is the unique positive real number with the property that $e^x$equals its own derivative.

*Proof.* Given $c^x$, its derivative (see the lemma) is $c^x \lim_{h\to 0} \frac{c^h-1}{h}$. However, comparing this to known derivative results from level 3, we see that

$$\lim_{h\to 0} \frac{c^h-1}{h} = ln(c)$$

In order for a number c to have the same property as e, we would need the derivative to be exactly $c^x$, so we would have to have $ln(c) = 1$, so $c = e$.

Note: if c is less than 1, then $c^{-x}$ is differentiable by the previous lemma and thus so is $c^x$.

□

## 5.4 The power rule

**Theorem.** $\frac{d}{dx}x^a = a * x^{a-1}$ if $a$ is a real constant. I emphasize - do NOT use this to differentiate something like $x^x$, this is only if $a$ is CONSTANT.

*Proof.*

$$\frac{d}{dx}x^a = \frac{d}{dx}e^{a*ln(x)} = e^{a*ln(x)}\frac{d}{dx}(a * ln(x)) = e^{a*ln(x)} * \frac{a}{x} = x^a * \frac{a}{x} = a * x^{a-1}$$

□

# 6 Integration

## 6.1 Riemann integration

What we want to do is make the idea that integration is "adding up the area of a bunch of thin rectangles" rigorous.

**Lemma.** (Bolzano Weierstrass theorem) Consider any infinite sequence which is bounded below and above, then you can extract an infinite subsequence of this sequence that converges to a limit.

*Proof.* We will plot the sequence on a graph and for each n consider the terms which are large enough that a larger term is never attained ever again. In figure 5, I show a picture of a graph of a generic graph with these points (which I will call peaks) circled.

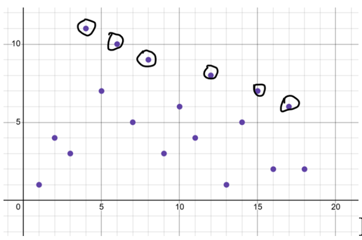

Figure 5

If there are infinitely many of these peaks, then these form an infinite non-increasing sequence which is bounded below, which has a highest lower bound that it must converge to (recall the discussion in the differentiation section). If there are finitely many peaks, this means there has to be an infinite sequence of non-decreasing points, since after the last peak, the next term is not a peak meaning there is a term after that term that is larger than it, and that term is also not a peak so there is a larger term after that and so on. This forms an infinite increasing sequence which is bounded above, which converges to its least upper bound.

□

If we have a function defined at 0, 1 and everywhere in between, then we want to say that the integration is the limit of a sum of areas of rectangles, kind of like in figure 6.

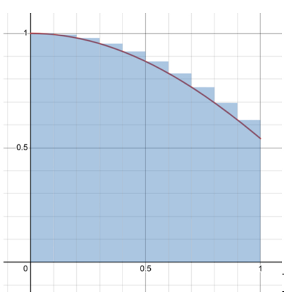

Figure 6

In mathematical notation, we want to say that the integral of $f(x)$ is

$$\lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n}f\left(\frac{r}{n}\right)$$

At first it may seem completely obvious that if f is continuous this limit will approach something. Where it is not obvious is if f is continuous but it is a fractal curve. You may reasonably say "but in A level we don't use fractal curves, so to justify A level we don't need to prove anything!", but the point is we still need some kind of mathematical proof, and it turns out that this is doable even if we merely assume continuity.

So, what I will do is consider the sums made by making the rectangles

i) just tall enough that they always overestimate the function (as in the image above). I will call this upper sum $U_n$if there are n rectangles.

ii) just short enough that they underestimate the function. I will call this lower sum $D_n$if there are n rectangles.

I will show that if $f$ is continuous these areas in fact get arbitrarily close together if I make n large enough.

Note that if a function is defined from a to b inclusive then the exact same argument applies because we can just rescale it appropriately.

**Theorem.** Continuous functions defined between 0 and 1 inclusive are integrable in the above sense.

*Proof.* Suppose on the contrary that there is an $\varepsilon > 0$ such that $U_n - D_n > \varepsilon$ for infinitely many n. For each such n, there must exist some interval, say the one from $\frac{k}{n}$ to $\frac{k+1}{n}$such that the range of values of the function on this interval is at least $\varepsilon$ wide, since if that were not the case we would have $U_n - D_n \leq \varepsilon$. It is therefore possible to pick an $x_n$and a $y_n$ both in this $\frac{k}{n}$ to $\frac{k+1}{n}$ interval such that $|f(x_n) - f(y_n)| > \varepsilon$ but clearly we have $|x_n - y_n| < \frac{1}{n}$.

Now the lemma comes in. We can extract a convergent subsequence of $x_n$which we will call $x_{n_k}$ (where $n_k$ is the corresponding increasing sequence of integers). This has a limit which we will call L. Since $x_n$ and $y_n$ get arbitrarily close together in the limit, we see that $y_{n_k}$ also tends to L. By continuity of f, we know that $f(x_{n_k}) \to f(L)$ and $f(y_{n_k}) \to f(L)$. But this is not possible, because it is always the case that $|f(x_{n_k}) - f(y_{n_k})| > \varepsilon$ so the idea that $f(x_{n_k}) - f(y_{n_k}) \to 0$ is a contradiction.

□

## 6.2 Fundamental theorem of calculus

**Theorem.** (Fundamental theorem of calculus, more formal version) Let f be a function with a continuous derivative between 0 and 1 inclusive which we can rescale necessarily, and call g the derivative of f. Then if $0 \leq t \leq 1$ then

$$\int\limits_0^t g(x)dx = f(t) - f(0)$$

where the integral is defined as above.

*Proof.* We want to make the level 3 intuition precise.

Clearly, $\int_0^t g(x)dx$ and $f(t) - f(0)$ are equal if $t = 0$, so we just have to show that for every t between 0 and 1, these have the same derivative. For any tolerance $\varepsilon > 0$, by continuity of $g$ we can find a $\delta$ small enough such that

$$\frac{\int_0^{t+\delta} g(x)dx - \int_0^t g(x)dx}{\delta} = \frac{\int_t^{t+\delta} g(x)dx}{\delta} = \frac{\int_t^{t+\delta} g(t) + \text{Error}dx}{\delta}$$

where $|\text{Error}| < \varepsilon$. Call the error term $h(x)$, then we have

$$\frac{\int_t^{t+\delta} g(t)dx + \int_t^{t+\delta} h(x)dx}{\delta} = g(t) + \frac{\int_t^{t+\delta} h(x)dx}{\delta}$$

In the second term, we are integrating a function on a $\delta$-wide interval, integrating it and dividing by $\delta$. This is the continuous analog of taking an average. Therefore, intuitively, we started with

$$\frac{\int_0^{t+\delta} g(x)dx - \int_0^t g(x)dx}{\delta}$$

and now we have

$$g(t) + A$$

where A is the mean value of h on our interval, which is clearly smaller than $\varepsilon$ since h is. Therefore, we can get as close as we like to $g(t)$ just by making $\delta$ sufficiently small, so we see that

$$\lim_{\delta \to 0} \frac{\int_0^{t+\delta} g(x)dx - \int_0^t g(x)dx}{\delta} = g(t)$$

So the right hand limit for the derivative exists. The left hand limit exists and is the same by considering $g(-x)$. Therefore, since the integral of g and f have the same derivative, and agree at $x = 0$, so they have to be equal everywhere.

□

### 6.3 Integration by substitution

We want to show that if u is some function of x with a continuous derivative, then

$$\int_{x=u(a)}^{u(b)} f(u(x))\frac{du}{dx}dx = \int_{u=a}^{b} f(u)du$$

As it turns out, the justification is exactly the same as what we will do to justify separation of variables in the next section.

## 7 Differential equations

**Theorem.** Suppose $\frac{dy}{dx} = \frac{f(x)}{g(y)}$ and f and g are continuous and suppose that the solution is such that on its domain of definition, g is never 0. Then$\int f(x)dx = \int g(y)dy$.

*Proof.* We know that $f(x) = g(y)\frac{dy}{dx}$, so $\int f(x)dx = \int g(y)\frac{dy}{dx}dx$.

We know the chain rule, so $\frac{d}{dx}\int g(y)dy = \frac{dy}{dx}\frac{d}{dy}\int g(y)dy = \frac{dy}{dx}g(y)$

On the other hand, $\frac{d}{dx}\int g(y)\frac{dy}{dx}dx = g(y)\frac{dy}{dx}$

Therefore, $\int g(y)\frac{dy}{dx}dx$ and $\int g(y)dy$ can only differ by a constant, so the result follows.

□

**Example.** If $\frac{dy}{dx} = \frac{1}{3}y^{-\frac{2}{3}}$and we impose the initial condition that the solution passes through $(0,0)$, something weird happens.

The normal method of separation of variables would give $y^{\frac{1}{3}} = x + c$ and then c is 0 by the initial condition so $y = x^3$, but we only have unique solutions subject to y not being 0 on the domain of definition. Therefore, if the domain of definition includes $x = 0$ where $y = 0$ automatically, we could have other solutions. Indeed we do, it turns out that the following function is a perfectly good solution. So we see that all the conditions of the theorem above were necessary.

$$\begin{cases} x^3 & x \leq 0 \\ 0 & 0 < x < 1 \\ (x-1)^3 & x \geq 1 \end{cases}$$

For example, if $y = x^3$ then $\frac{dy}{dx} = 3x^2 = 3y^{\frac{2}{3}}$ but if $y = 0$ then $\frac{dy}{dx} = 0 = 3y^{\frac{2}{3}}$, and the derivative is 0 at the boundary points. So indeed, the equation is satisfied.

We want to show that if we have a solution that is above 0 (or more generally, does not cross the region where we would have divided by 0 during the separation of variables process) on some domain we cannot have a solution that crosses 0, so it is actually unique. I will try to sidestep the Analysis approach since that is too complicated for this level. I will go through an example to explain the rationale.

We want to show that it is ok to say that $\frac{dy}{dx} = y$ and passes through $(0,1)$ implies that $\frac{1}{y}\frac{dy}{dx} = 1$ and we can continue and get $y = e^x$ as a unique solution, for the reason that if we did have some bad solution that passed through $y = 0$ (which is the bad point), we would have a contradiction.

Clearly, a solution has a derivative everywhere and is therefore continuous. Lets say we had a solution that crossed 0 when $x = -3$, then we could decide that the domain is $x > -2.9999$, or something so close to 3 that the solution is actually less than $e^{-3}$ somewhere in this domain. We can get close enough since the solution is continuous. Then we would have two genuinely different solutions where y is not 0, contradicting the above theorem.

# 8 Geometric series

**Theorem.** Suppose we have a geometric series with initial term a common ratio r, then the formula $\frac{a}{1-r}$ is valid if $|r| < 1$ and invalid if $|r| > 1$.

*Proof.* Before reading on, try using your calculator to calculate the sum of the first few terms of a geometric series with common ratio

(i) 0.5

(ii) 1.5

See how the values change as you add more terms. Can you figure out why we need that for the infinite sum the common ratio has to be between -1 and 1?

The partial sums of a geometric series are given as follows, where the first term is a and the common ratio is r and the sum of the first k terms is

$$\frac{a(1-r^k)}{1-r}$$

This is a fact known from level 3. Now note that as k goes to infinity, if $|r| < 1$ then $r^k \to 0$ so the expression above actually approaches $\frac{a}{1-r}$ and conversely if $|r| > 1$ then the expression above will blow up to infinity as k increases and not approach a limit. If you're not sure why, consider what happens if you repeatedly multiply by

(i) something less than 1

(ii) something more than 1

(iii) the fact that if it's something negative you just have an alternating sign and that doesn't affect convergence

If r is exactly 1 we are just adding the same thing and it will go to infinity. If r is -1 we will oscillate between 2 values forever and not approach a limit.

□

# 9 Surface areas and volume

Most of the stuff was proved in level 3. We are just missing one.

**Proposition.** The surface area of the curved part of a cone is $\pi rl$ where $l$ means the distance if you measure from the base to the tip of the cone.

*Proof.* Imagine taking the surface of a cone and cutting and rolling it as shown in figure 7.

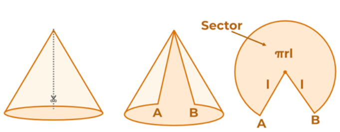

Figure 7

Then the distance from the point at the top (the center of the resulting circular arc) to A and B is $l$, so $l$ is the radius of the resulting circle. The circumference of the circle at the bottom of the cone is $2\pi r$ and hence $2\pi r$ is the curved arc length of the resulting circular arc. Therefore, the angle in radians traced out by the resulting circular arc is $\frac{2\pi r}{l}$. Hence, by the formula for the area of a sector, we just need $\frac{1}{2}l^2\frac{2\pi r}{l} = \pi rl$.

□

## 10 Introduction to complex numbers

We are doing this because we need it for the next section.

So basically what you need to know is that i is an imaginary number defined by $i = \sqrt{-1}$, and we often draw numbers of the form $a + bi$ as the point $(a, b)$ in a 2D grid (This 2D grid is called an argand diagram). Now we consider a very important geometric intuition about multiplying complex numbers.

In order to understand this, you should either try to visualize or draw this until it makes sense to you. 3blue1brown's lockdown math explains all these concepts beautifully, but I cannot guarantee that those will exist since I am not affiliated with 3blue1brown nor youtube.

**Definition.** Given a complex number $z$ in the argand diagram, $|z|$ is its distance from the origin and $\arg(z)$ is the angle it makes with the real axis in radians, where we define it by convention so that this is between 0 and $\pi$ if we are at or above the real axis and between $-\pi$ and 0 otherwise.

When you multiply 2 complex numbers $a * b$, what you can think of is that you stretch and rotate the entire argand diagram such that the number 1 lines up with the number a, then go to the number b on that new argand diagram. Note that $i(a + bi) = ai - b$ by algebra. Draw $a + bi$ and $ai - b$ on an argand diagram and try to convince yourself that they are always perpendicular. Now the reason my trick from earlier works is because if you have $a(x + yi)$ this is $ax + ayi$. Because moving on path 1 then path 2 on an argand diagram is the same as adding the numbers given by path 1 and path 2, we have that if you move |a|x units in the direction parallel to $a$ then $|a|y$ units in the direction perpendicular to $a$. It should hopefully be clear that this gets you to $a(x + yi)$ in the normal grid and $x + yi$ in the grid stretched and rotated (specifically, stretched by a factor of $|a|$ and rotated by $arg(a)$ radians).

Below is an illustration for both the multiplication idea with the example $(3 + 4i)(2 + i)$ and in the case of multiplying on the unit circle. (Grids not accurately drawn)

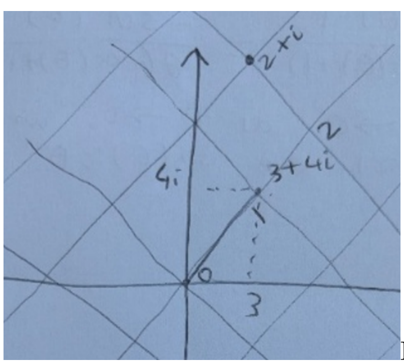

Figure 8

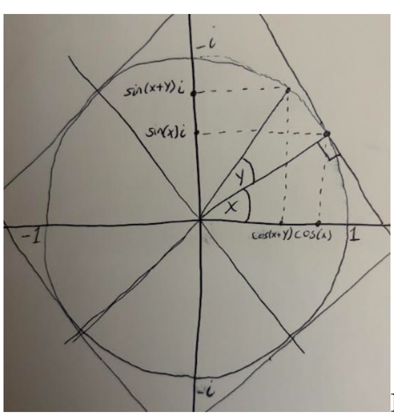

Figure 9

Figures 8 and 9 show rotated and stretched grids on an argand diagram to visually show my point, with unit circles and stuff included in the diagrams.

Figure 9 shows the case where we are multiplying things that are on the unit circle with arguments x and y, which geometrically are given by $\cos(x) + i\sin(x)$ and $\cos(y) + i\sin(y)$, since cos and sin are coordinates along the unit circle.

As an extra challenge for capable/interested people, can you use these ideas to find that $\pm\frac{1}{2}\sqrt{2}(1+i)$ are two square roots of i and that two other cube roots of 1 that are not just 1 are given by $-\frac{1}{2} \pm \frac{1}{2}\sqrt{3}i$ without directly computing the squares or cubes but rather using the ideas above and known sines and cosines of certain angles? This is just an exercise and these ideas will be covered later level 5 and are covered in the previously mentioned lockdown math youtube videos anyway.

# 11 Trigonometry

## 11.1 Addition formulae

**Theorem.** We have the following formulae:

i) $sin(x + y) = sin(x)cos(y) + cos(x)sin(y)$

ii) $cos(x + y) = cos(x)cos(y) - sin(x)sin(y)$

*Proof.* Some textbooks give a geometric proof, but this proof justifies it for all angles, not just angles between 0 and 90 degrees, and is much more beautiful.

Consider the function $cos(x) + isin(x)$ with $x$ real. On an Argand diagram, this is the function you get from walking $x$ radians around the unit circle anticlockwise, since on an argand diagram if I have the function you get from walking $x$ radians around the unit circle then indeed you can see that $cos(x)$ will be the real part and $sin(x)$ will be the imaginary part. You can draw a diagram to convince yourself of this.

Now consider $(cos(x)+isin(x))(cos(y)+isin(y))$ with $x, y$ real. Using the multiplication trick from earlier, what you should do is rotate the argand diagram $x$radians so that the point 1 on your rotated grid corresponds to the point $cos(x)+isin(x)$ on your original argand diagram. Then you can see that the point $cos(y) + isin(y)$ on your rotated diagram is gotten to by walking form the original point 1 $x$ radians anticlockwise then $y$ radians anticlockwise, so you walk $x + y$ radians anticlockwise. This is significant because you then end up at the point $cos(x+y)+isin(x+y)$, which intuitively establishes the following identity:

$$(cos(x) + isin(x))(cos(y) + isin(y)) = cos(x + y) + isin(x + y)$$

Notice how this behavior is similar to exponentials. We will expand on this idea in the exponentials and logarithms video.

Now expand the left hand side, noting that $i^2 = -1$ by definition, then we have

$$cos(x)cos(y) - sin(x)sin(y) + i(cos(x)sin(y) + sin(x)cos(y) = cos(x + y) + isin(x + y)$$

Equating the real and imaginary parts to eachother gives the desired result.

This proof will never not be beautiful to me.

□

## 11.2 Limits

**Theorem.** We have the following limits which we used to compute the derivatives of sin and cos:

i) $\lim_{\theta\to 0}\left(\frac{sin(\theta)}{\theta}\right) = 1$

ii) $\lim_{\theta\to 0}\left(\frac{cos(\theta)-1}{\theta}\right) = 0$

*Proof.* These are best demonstrated with a picture. See figure 10.

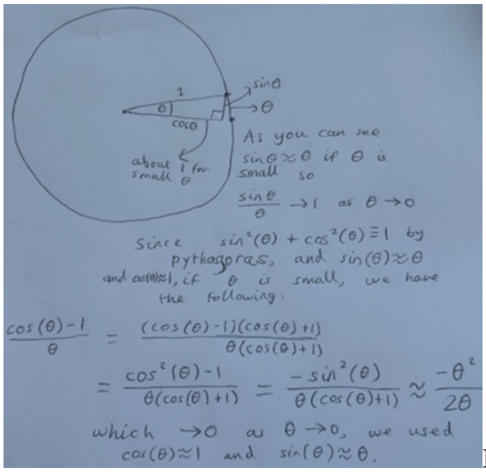

Figure 10

□

## 12 Dot products

**Definition.** (Basis vectors) Define i to be the vector with length 1 that points in the x direction (Think of this as to the right). Define j to be the vector with length 1 that points in the y direction (Think of this as to the back). Define k to be the vector with length 1 that points in the z direction (Think of this as up). These are the basis vectors, in particular every vector is given by a list of what its components are in the directions of these vectors. This idea is useful to keep in mind.

**Theorem.** The "multiply components and add the results" and "multiply the lengths and multiply by the cosine" interpretations of the dot product are the same.

*Proof.* I will prove this in 2 ways.

First, start from the cosine definition and consider figure 11.

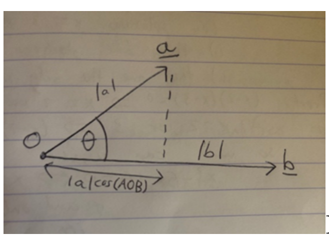

Figure 11

Now hopefully you can see that if I add a vector c to the vector a, then the amount that the length of its projection in the direction of b will change by will be equal to the length of the projection of c onto b (which will be negative if c is facing away from b).

So hopefully you can see visually that

$$|a|cos(AOB) + |c|cos(COB) = |a + c|cos((A + C)OB)$$

Multiplying both sides by $|b|$ and then substituting in the dot proudct definition gives the important identity $a.b + c.b = (a + c).b$. Since both definitions of the dot product are the same if you swap the order of the two components around, we have $b.a + b.c = b.(a + c)$.

Now, observe that the basis vectors $i.i, j.j, k.k$ are all equal to 1 because, eg $i.i = |i||i|cos(0) = 1$. However, $i.j, i.k$ and $k.j$ are all equal to 0 because, eg, i is perpendicular to j so the cosine of the angle between them is 0.

Now, lets say for example that v and u are 3 dimensional vectors and we want to compute $v.u$. We can write both v and u as $ai + bj + ck$ and $di + ej + fk$ respectively for some numbers $a, b, c, d, e, f$.

Then

$$v.u = (ai + bj + ck).(di + ej + fk) = (ai).(di + ej + fk) + (bj).(di + ej + fk) + (ck).(di + ej + fk)$$

where we have used the fact that we can split the dot product up like this as we just demonstrated, so

$$v.u = ai.di + ai.ej + ai.fk + bj.di + bj.ej + bj.fk + ck.di + ck.ej + ck.fk$$

$$v.u = ad(i.i) + ae(i.j) + af(i.k) + bd(j.i) + be(j.j) + bf(j.k) + cd(k.i) + ce(k.j) + cf(k.k)$$

Using the fact that $i.i, j.j, k.k = 1$ and everything else is 0, we get

$v.u = ad + be + cf$, so we can indeed derive the algebraic definition from the geometric definition.

The second proof, on the other hand, derives the geometric definition from the algebraic definition.

The second proof starts by assuming that we can in fact get the dot product by summing component-wise, then derives $a.b = |a||b|cos(AOB)$. Start with the fact that $a.a$ is the sum of the squares of the components of a, which by pythagoras is in fact equal to the square of the length of a. But if we have a triangle ABC, then by the cosine rule we know that

$$|\overrightarrow{BC}|^2 = |\overrightarrow{AC}|^2 + |\overrightarrow{AB}|^2 - 2|\overrightarrow{AC}||\overrightarrow{AB}|cos(BAC) \tag{2}$$

Now we can do some algebra on this:

$$|\overrightarrow{AC}|^2 + |\overrightarrow{AB}|^2 - 2|\overrightarrow{AC}||\overrightarrow{AB}|cos(BAC) = \overrightarrow{AC}.\overrightarrow{AC} + \overrightarrow{AB}.\overrightarrow{AB} - 2|\overrightarrow{AC}||\overrightarrow{AB}|cos(BAC)$$

$$|\overrightarrow{BC}|^2 = |\overrightarrow{AC} - \overrightarrow{AB}|^2 = (\overrightarrow{AC} - \overrightarrow{AB}).(\overrightarrow{AC} - \overrightarrow{AB}) = \overrightarrow{AC}.\overrightarrow{AC} + \overrightarrow{AB}.\overrightarrow{AB} - 2\overrightarrow{AC}.\overrightarrow{AB}$$

By equation (2), everything in the above 2 equations is equal to eachother. Therefore,

$$\overrightarrow{AC}.\overrightarrow{AC} + \overrightarrow{AB}.\overrightarrow{AB} - 2|\overrightarrow{AC}||\overrightarrow{AB}|cos(BAC) = \overrightarrow{AC}.\overrightarrow{AC} + \overrightarrow{AB}.\overrightarrow{AB} - 2\overrightarrow{AC}.\overrightarrow{AB}$$

Cancelling common terms and common factors gives

$$|\overrightarrow{AC}||\overrightarrow{AB}|cos(BAC) = \overrightarrow{AC}.\overrightarrow{AB}$$

as required.

□

*Remark.* Note that we can move numbers outside of dot products like this: $av.bu = ab(v.u)$. This is immediate from the algebraic definition.

# 13 The generalized binomial theorem

## 13.1 Introduction to taylor series

The generalized binomial theorem is a special case of a concept called a Taylor series. A Taylor series is essentially when we have a function and we use the following method to try to fit an infinite polynomial to the function $f(x)$: Let the constant term of the polynomial be $f(0)$ so the functions and our polynomial at least match in terms of their value at $x = 0$. Now we let the coefficient of $x$ in our polynomial be $f'(0)$ so our polynomial and its slope at $x = 0$ match that of the original function, so if we zoom into the graph enough that $f$ looks like roughly a straight line (since we are assuming $f$ is "smooth" using the intuitive informal definition) our polynomial will be a decent approximation for $f$. Now we let the term in $x^2$ of our polynomial be $\frac{1}{2}f''(0)$ so that the second derivatives at $x = 0$ match. We keep doing this until we get an infinite polynomial, and the hope is that this polynomial well approximates the original function as we add more terms, and that in the limit the infinite polynomial we get equals the original function. The infinite polynomial we get is called the Taylor series of the function. This is true in a lot of cases, and we prove it for the binomial theorem $(1 + x)^n$ as that is used at A level, as the binomial expansion is actually the Taylor series of $(1 + x)^n$ for general n.

However, we do necessarily need to center the series at $x = 0$. The general form of a taylor series is

$f(x_0) + (x - x_0)f'(x_0) + \frac{(x-x_0)^2}{2!}f''(x_0) + \frac{(x-x_0)^3}{3!}f'''(x_0) + ... + \frac{(x-x_0)^n}{n!}f^{(n)}(x_0) + ...$

It is not at all obvious that this series actually converges. It is also not obvious that this (if you add up infinitely many terms) actually equals $f(x)$ and is not just a good approximation, indeed you can only sometimes conclude this, as we will do with the series for $(1+x)^n$, which we will demonstrate is convergent and equal to the function when $|x| < 1$ and does not converge at all if $|x| > 1$.

## 13.2 The proof

**Fact.** We will justify this in the appendix, but under certain conditions (which are satisfied in the following proof), the derivative of a power series is the sum of the derivatives of each term.

**Proposition.** This fact is not completely obvious and actually needs proof.

*Proof.* Consider the series

$$f(x) = \sin(x) + \left(\frac{1}{2}\sin(2x) - \sin(x)\right) + \left(\frac{1}{3}\sin(3x) - \frac{1}{2}\sin(2x)\right) + \left(\frac{1}{4}\sin(4x) - \frac{1}{3}\sin(3x)\right) + ...$$

We see that if we stop the series after n terms, we have $f(x) = \frac{1}{n}\sin(nx)$ which has $|f(x)| \leq \frac{1}{n}$. Therefore, the limit of these finite sums, which is the definition of the infinite sum, is 0. Now suppose we cuold differentiate this term by term, we would get

$$f'(x) = \cos(x) + (\cos(2x) - \cos(x)) + (\cos(3x) - \cos(2x)) + (\cos(4x) - \cos(3x)) + ...$$

If we stop *this* after n terms, we have $\cos(nx)$, which clearly does not tend to 0, so it is not consistent.

□

**Theorem.** The taylor series for $(1+x)^n$, which we will demonstrate is convergent and equal to the function when $|x| < 1$ and does not converge at all if $|x| > 1$. Furthermore, this taylor series is actually the same as the binomial series which we stated in level 3.

*Proof.* Using section 7 and level 3 techniques, we see that on the domain of numbers between -1 and 1 non-inclusive, the unique solution to the following differential equation subject to the condition that it passes through the point $(0, 1)$

$$\frac{dy}{dx} = \frac{ny}{1+x}$$

is $y = (1+x)^n$. By uniqueness and the fact above, the theorem will follow if we can show that the series

$$y = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + ...$$

*also* satisfies the differential equation. Since if it does, we conclude that the 2 versions of y must in fact be equal to eachother.

Let y be the power series definition, then we know how to differentiate, here is how that goes:

$$\frac{dy}{dx} = n + n(n-1)x + \frac{n(n-1)(n-2)}{2!}x^2 + \frac{n(n-1)(n-2)(n-3)}{3!}x^3 + ...$$

$$x\frac{dy}{dx} = nx + n(n-1)x^2 + \frac{n(n-1)(n-2)}{2!}x^3 + \frac{n(n-1)(n-2)(n-3)}{3!}x^4 + ...$$

$$(1+x)\frac{dy}{dx} = \frac{dy}{dx} = n + n(n-1)x + \frac{n(n-1)(n-2)}{2!}x^2 + \frac{n(n-1)(n-2)(n-3)}{3!}x^3 + ... + nx + n(n-1)x^2 + \frac{n(n-1)(n-2)}{2!}x^3 + \frac{n(n-1)(n-2)(n-3)}{3!}x^4 + ...$$

Don't worry, this simplifies nicely.

$$(1+x)\frac{dy}{dx}$$
$$= n+nx(1+(n-1))+n(n-1)x^2\left(1+\frac{(n-2)}{2}\right)+\frac{n(n-1)(n-2)}{2!}x^3\left(1+\frac{(n-3)}{3}\right)+\frac{n(n-1)(n-2)(n-3)}{3!}x^4\left(1+\frac{(n-4)}{4}\right)+\cdots$$
$$= n+nx(n)+n(n-1)x^2\left(\frac{n}{2}\right)+\frac{n(n-1)(n-2)}{2!}x^3\left(\frac{n}{3}\right)+\frac{n(n-1)(n-2)(n-3)}{3!}x^4\left(\frac{n}{4}\right)+\cdots$$
$$= n\left(1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+...\right)$$
$$= ny$$

Therefore the differential equation is satisfied. We have $(1+x)\frac{dy}{dx} = ny$ which is just a rearrangement of the original thing.

Unfortunately, we are not done since we need to address why this is valid whenever $|x| < 1$. It is basically because if x is too big the terms get too big for the series to converge to a value.

Consider what the ratio is between, for example, the terms in $x^{99}$and $x^{100}$. n is fixed so to get from the 99 one to the 100 one, we need to divide by 100 and multiply by $x(n-99)$. So the ratio is $\frac{(n-99)x}{100}$. When this "100" gets so big that it is far bigger than n, thise will approach $x$. In particular, if $|x|$ is less than 1, the ratio after some time will eventually be always less than 1, and we saw in section 8 that this implies convergence. Similarly, if $|x| > 1$ the series diverges.

The exception is when n is an integer, in which case there is a 0 term so all subsequent terms become 0 regardless of the ratio.

□

### 13.3 Other taylor series

Here are other taylor series for functions and the differential equation you would use to argue that they are valid.

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + ...$, convergent everywhere since the ratio goes to 0, proven using the differential equation $\frac{dy}{dx} = y$ subject to passing through $(0, 1)$.

$ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$, convergent if $|x| < 1$, comes from reverse-differentiating the series for $\frac{1}{1+x}$.

In the video on exponentials and logarithms we will see that $e^{ix} = cos(x) + i * sin(x)$, which is Euler's famous identity. Writing out the taylor series for the left hand side using the one for $e^x$ gives

$$cos(x) + i * sin(x) = e^{ix} = 1 + ix - \frac{x^2}{2!} - i\frac{x^3}{3!} + \frac{x^4}{4!} + i\frac{x^5}{5!} - \frac{x^6}{6!} - i\frac{x^7}{7!} + \frac{x^8}{8!} + ...$$

Equating the real and imaginary parts gives the following series for cos and sin, convergent and valid everywhere:

$$cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ...$$
$$sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ...$$

This gives us an alternative way to see the limits from section 11.

## 14 Partial fractions

This is surprisingly tricky. If the explanation of "you can just verify that your solution in the exam works", you can skip this section. However, I really do like knowing where the general rule actually comes from.

**Lemma.** (Bezout's identity): Let $P_1(x)$ and $P_2(x)$ be polynomials which have no factors in common. Then there exist polynomials A and B such that $AP_1(x) + BP_2(x) = 1$.

*Proof.* Lets rename the polynomials if necessary so that $P_1$ is the one with larger degree. If their degrees are the same it does not matter.

Now do the normal long division process to get $P_1 = Q_1P_2 + R_1$ where $deg(R_1) < deg(P_2)$. Now we work with polynomials $R_1$ and $P_2$ and do the same thing, except now the smallest degree of the two is lower. We get $P_2 = Q_2R_1 + R_2$ with $deg(R_2) < deg(R_1)$. The remainder of the proof will mostly be a demonstration using an example.

**Example.** Suppose that

$$P_1 = (x-1)^2(x-2) = x^3 - 4x^2 + 5x - 2$$

$$P_2 = (x+1)(x+3) = x^2 + 4x + 3$$

Our goal is to find polynomials A and B such that

$$AP_1 + BP_2 = 1$$

We first try to do the long division process to get $P_1 = Q_1P_2 + R_1$. If we do this, we will get

$$P_1 = (x-8)P_2 + (34x - 22)$$

So $R_1 = 34x - 22$. Also,

$$R_1 = P_1 - (x-8)P_2$$

The next step is to try to divide $P_2$ by $R_1$ using polynomial long division, and call the remainder $R_2$. We get

$$P_2 = \left(\frac{1}{34}x + \frac{57}{578}\right)R_1 + \frac{240}{289}$$

This means that $R_2 = \frac{240}{289}$ and

$$\frac{240}{289} = P_2 - \left(\frac{1}{34}x + \frac{57}{578}\right)R_1$$

At this point we have reached a constant. We may reach a constant at $R_3$ or later but the point is to repeat this procedure to get smaller remainders.

$$1 = \frac{289}{240}P_2 - \frac{289}{240}\left(\frac{1}{34}x + \frac{57}{578}\right)R_1$$

But remember we had written $R_1$ as something times $P_1$ plus something times $P_2$. By doing the same procedure, we can always write $R_2$ in this way. But then we can try to divide $R_1$ by $R_2$ to get a remainder $R_3$ with degree less than $R_2$. But if $R_1 = Q_3R_2 + R_3$ then $R_3 = R_1 - Q_3R_2$, and since $R_1$ and $R_2$ can be written in this way, so can $R_3$. We continue by induction.

However, we are not done. If you have been paying attention, you would notice that we have not used the fact that $P_1$ and $P_2$ have no factors in common. It turns out that the reason we need this condition is so that one of the remainder terms is actually a non-zero constant like the $\frac{240}{289}$ in the example above.

Any two pairs of polynomials we work with will not share any common factors.

Reason: lets say for the sake of example that $R_3$ and $R_4$ shared a common factor, then $R_2$ which is a quotient polynomial times $R_3$ plus $R_4$ will have that factor, then $R_1$ which is a quotient polynomial times $R_2$ plus $R_3$ must also have that factor, and we keep going until realizing that by this reasoning our original $P$'s must have both had that factor, and thus a factor in common.

Therefore, one of the remainders is indeed a constant because if $R_n$ was a polynomial of degree at least 1 but then $R_{n+1}$ was suddenly 0 and not a non-zero constant, then that would mean that $R_n$ would divide $R_{n-1}$ with zero remainder as we would get$R_n = QR_{n-1} + 0$, meaning $R_n$ and $R_{n-1}$ share a common factor of $R_{n-1}$ which is a contradiction, as our original $P$'s were assumed to have no common factors.

□

*Remark.* An analagous proof shows that if two **numbers** x and y have no common prime factors, then you can find numbers a and b such that $ax + by = 1$. This is a very useful result in number theory.

**Theorem.** The general rule for partial fractions from A levels always works. Specifically, if we have a fraction like

$$\frac{P(x)}{(x-a_1)^{m_1}(x-a_2)^{m_2}...(x-a_n)^{m_n}}$$

where $P(x)$ is a polynomial, then this fraction can be rewritten in the following form:

$$Q(x)+\left[\frac{b_1}{x-a_1}+\frac{b_2}{(x-a_1)^2}+...+\frac{b_{m_1}}{(x-a_1)^{m_1}}\right]+\left[\frac{c_1}{x-a_2}+\frac{c_2}{(x-a_2)^2}+...+\frac{c_{m_2}}{(x-a_2)^{m_2}}\right]+...+\left[\frac{d_1}{x-a_n}+\frac{b_2}{(x-a_n)^2}+...+\frac{b_{m_n}}{(x-a_n)^{m_n}}\right]$$

where $Q(x)$ is a polynomial.

Note that this looks intimidating because it is **very** general. In practice, the m's and n's will very rarely be greater than 2 or 3.

*Proof.* $Q(x)$ just comes from the long division, so we will assume we just have a remainder term, ie the degree of the numerator is smaller than the degree of the denominator. Lets call the numerator F and the demoninator G. Our goal is to manipulate $\frac{F}{G}$.

Define

$$G_1 := (x-a_1)^{m_1}$$

$$G_2 := (x-a_2)^{m_2}(x-a_3)^{m_3}...(x-a_n)^{m_n}$$

Then by the previous lemma, it is possible to find polynomials C and D such that $CG_1 + DG_2 = 1$.

$$\frac{F}{G} = \frac{F(CG_1 + DG_2)}{G_1G_2} = \frac{DF}{G_1} + \frac{CF}{G_2}$$

Now do the following long division ($DF$ by $G_1$) to get

$$DF = G_1Q + F_1$$

with $Deg(F_1) < Deg(G_1)$, and define

$$F_2 := CF + QG_2$$

We then get

$$\frac{F}{G} = \frac{F_1 + G_1Q}{G_1} + \frac{F_2 - G_2Q}{G_2} = \frac{F_1}{G_1} + Q + \frac{F_2}{G_2} - Q = \frac{F_1}{G_1} + \frac{F_2}{G_2}$$

Multiplying both sides by $G = G_1G_2$ gives

$$F = F_2G_1 + F_1G_2$$

Thus,

$$deg(F_2) = deg(F - F_1G_2) - deg(G_1) \leq max(deg(F), deg(F_1G_2)) - deg(G_1) < max(deg(G), deg(G_1G_2)) - deg(G_1) = deg(G_2)$$

Here is where this line actually comes from:

First step - The degree of the product of polynomials is the sum of the degrees so $deg(F - F_1G_2) = deg(F_2) + deg(G_1)$ since $F = F_2G_1 + F_1G_2$.

Second step: The difference between two polynomials can't be greater than the degree of both of them.

Third step: We know that $deg(F) < deg(G), deg(F_1) < deg(G_1)$

Fourth step: Since $G = G_1G_2$ the max term is just $deg(G_1G_2)$ so the conclusion follows.

We can now do the same thing we did to $\frac{F}{G}$ to $\frac{F_2}{G_2}$ and continue until we run out of roots of G. We will get

$$\frac{F}{G} = \frac{F_1}{(x-a_1)^{m_1}} + \frac{F_2}{(x-a_2)^{m_2}} + \frac{F_3}{(x-a_3)^{m_3}} + ... + \frac{F_n}{(x-a_n)^{m_n}}$$

Where the degree of each numerator is less than the degree of the corresponding denominator.

Now for each term (I will demonstrate with the first one), we can do the following trick to get this into the desired form:

Set $y = x - a_1$ and set $H_1(y) = F_1(y + a_1)$ which is clearly still a polynomial with degree $< m_1$, then write

$$\frac{F_1(x)}{(x-a_1)^{m_1}} = \frac{F_1(y+a_1)}{(y)^{m_1}} = \frac{d_0 + d_1 y + d_2 y^2 + ... + d_{m_1-1} y^{m_1-1}}{y^{m_1}}$$

where the d's are (possibly zero constants). Now, this is just

$$\sum_{i=0}^{m_1-1} \frac{d_i}{y^{n-i}}$$

So it follows that the term can be written like $\left[\frac{b_1}{x-a_1} + \frac{b_2}{(x-a_1)^2} + ... + \frac{b_{m_1}}{(x-a_1)^{m_1}}\right]$exactly as we claimed at the start. Since we can do this for all n terms, the result follows.

□

*Remark.* It turns out that *every* polynomial can be decomposed into a form like $(x-a_1)^{m_1}(x-a_2)^{m_2}...(x-a_n)^{m_n}$ if you allow the a's to be complex. This is called the fundamental theorem of algebra, but we will not prove it in this level (we don't need it).

## 15 Inverses of differentiable functions

We want to show that if we can differentiate a function we can differentiate its inverse, and therefore conclude that we can differentiate stuff like arcsin and log. In level 3 we took it as obvious, but here we will prove it properly.

Again, if looking at the graph satisfies you, you can skip this section.

**Theorem.** Let a<b and let f be a function of x which is continuous between a and b inclusive and differentiable between a and b non-inclusive. Suppose that the derivative is always positive or always negative, then the inverse of f is differentiable except possibly at its end points. Furthermore, the derivative of $f^{-1}$ is the reciprocal of the derivative of $f$ at the corresponding point (Intuitively, $\frac{dy}{dx} = 1/\frac{dx}{dy}$).

*Proof.* We will take it as given that the inverse is continuous (of course you can still draw it without lifting your pen just reflecting about $y = x$). Proving this rigorously is for Analysis, which is essentially the nuclear option when our goal is to just illuminate where A level techniques come from.

Since the derivative is always positive or always negative, the function is increasing or decreasing and therefore one-to-one, so it actually has a proper inverse. Given y between $f(a)$ and $f(b)$ and not equal to either of these, and h small enough that $y + h$ and $y - h$ are still between them, define $x = f^{-1}(y)$ and $k = f^{-1}(y+h) - x$. Then consider

$$\frac{f^{-1}(y+h) - f(y)}{h}$$

If this has a limit as h goes to 0, we would be done. Indeed, this is just (by definition)

$$\frac{k}{f(x+k) - f(x)}$$

The limit of this as k goes to 0 exists and is the reciprocal thing we want, and it is never 0 since f is increasing or decreasing so we can take the reciprocal. But by continuity, as k goes to 0 h also goes to 0. Hence, the result follows.

□

## 16 Cobweb diagrams and newton raphson

There is a video on this on the website.

## 17 Exponentials and logarithms

There is a video on this on the website. Sections 5, 7, 10 and 15 are prerequesite.

## 18 (Appendix A) Motivation for the exponentials and logarithms video

Consider the following question: Suppose we have the differential equation

$$\frac{dy}{dx} = y$$

By separation of variables, we get

$$ln(y) = x + c$$

Therefore

$$y = e^x e^c$$

But notice, that $y = -e^x$ is a genuine solution, but there is no real c such that $e^c = -1$ (look at the graph!)

The idea is it works if c is a complex number. We will discuss this point in the video.

The video derives from basic exponent laws what it means to take a positive real number to a non-integer real power and why this definition is internally consistent, and then defines exponentiation with complex exponents and extend the definition of powers to the complex numbers. I define the complex logarithm and the problems that come from it being multi-valued or having a branch cut. I explain which exponent and logarithm rules work in the complex numbers and why. I then discuss a bit about the integral of $1/x$ and give a better explanation for what the deal is with $ln(x)$ vs $ln(|x|)$. I do a brief aside on contour integration (Don't worry if you don't understand this, I explain it again in levels 6 and 8 with increasing detail in each level.) since it will be used in a later proof, then finally go back and answer the motivating differential equations question above.

## 19 (Appendix B) Technical details from section 13

When we justify differentiating a power series we will be working with sums. We note that intuitively, we know that if we rearrange the order of the terms in the sum, the sum will not change. However, this is not always true for infinite sums.

**Proposition.** The idea that we can reorder terms in an infinite sum is actually something that requires justification

*Proof.* Suppose for a contradiction that I'm being silly and it always works because of course it does, obviously.

Consider the infinite series

$$S := 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - ...$$

Then

$$2S = 2 - 1 + \frac{2}{3} - \frac{2}{4} + \frac{2}{5} - \frac{2}{6} + \frac{2}{7} - ... = 1 + \frac{2}{3} - \frac{1}{2} + \frac{2}{5} - \frac{1}{3} + \frac{2}{7} - \frac{1}{4} + ...$$

Notice that the terms with odd denominator appear twice but the terms with a positive denominator appear once. We can rearrange to get

$$2S = 1 - \frac{1}{2} + \left(\frac{2}{3} - \frac{1}{3}\right) - \frac{1}{4} + \left(\frac{2}{5} - \frac{1}{5}\right) - \frac{1}{6} + \left(\frac{2}{7} - \frac{1}{7}\right) - ...$$
$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - ... = S$$

Therefore $2S = S$, so we would conclude $S = 0$. However, it is easy to see that after the second term (where the sum is $\frac{1}{2}$) we never go below $\frac{1}{2}$ again since we increase then decrease by something smaller over and over again, so the limit cannot be 0. Contradiction.

$\square$

Fun fact: It turns out that the actual value of S is $ln(2)$. Although it requires further justification since it is on the edge of the "valid interval", it is to do with the taylor series for log.

(Triangle inequality) Let $a_1, a_2, ..., a_n$ be real or complex numbers. Then

$$|a_1 + a_2 + a_3 + ... + a_n| \leq |a_1| + |a_2| + |a_3| + ... + |a_n|$$

*Proof.* because the first expression is like walking a distance of $|a_1|$ in some direction in the complex plane, then a distance of $|a_2|$ in some other direction, then a distance of $|a_3|$ in some other direction, and so on, then recording your distance from the origin at the end. The right hand expression is like walking all of those distances in the same direction and then recording how far you've walked, which intuitively will always be a distance at least as long ("line is the shortest distance between 2 points").

$\square$

**Definition.** A sum $\sum_{n=1}^{\infty} a_n$ is said to "converge absolutely" if $\sum_{n=1}^{\infty} |a_n|$ converges.

**Lemma.** The triangle inequality holds for infinite sums

*Proof.* Since it holds for all finite sums, the difference

$$|a_1| + |a_2| + |a_3| + ... + |a_n| - |a_1 + a_2 + a_3 + ... + a_n|$$

cannot tend to a negative limit, which it would have to for the lemma to be false.

$\square$

**Theorem.** If a sum converges absolutely, rearranging or regrouping the terms will not affect the result.

*Proof.* Let S be $\sum_{n=1}^{\infty} a_n$ and L be $\sum_{n=1}^{\infty} |a_n|$, then by convergence, for any tolerance $\varepsilon$, no matter how small, we can pick M large enough such that $\sum_{n=M+1}^{\infty} |a_n| < \varepsilon$.

Let $b_n$ be a reordering of the $a_n$'s and define $S_k := \sum_{n=1}^{k} b_n$. We want to show that as n goes to infinity, $S_k - S \to 0$. Equivalently, we want to show that for any $\varepsilon$ it is possible to pick N large enough such that $|S_N - S| < \varepsilon$.

Pick M as above then let $C$ be so large that in $b_1, b_2, b_3, ..., b_C$ the terms $a_1, a_2, ..., a_M$ all appear. Then

$$S_C = \sum_{N=1}^{C} b_n = \sum_{n=1}^{M} a_n + \sum_{n>M \text{ and } a_n \text{ is in the first C b terms}} a_n$$

Now

$$|S_C - S| = \left| S_C - \sum_{n=1}^{\infty} a_n \right| = \left| \sum_{n>M \text{ and } a_n \text{ is in the first C b terms}} a_n - \sum_{n=M+1}^{\infty} a_n \right|$$

$$= \left| \sum_{n>M \text{ and } a_n \text{ is NOT in the first C b terms}} a_n \right| \leq \sum_{n=M+1}^{\infty} |a_n| < \varepsilon$$

The second last inequality is by the triangle inequality and the fact that if we add in extra terms it can only increase it. Therefore $|S_C - S| < \varepsilon$ as we wanted to show.

So we can rearrange the terms. Now we want to show that we can regroup them, for example, we want to know that we can write

$$x + x^2 + x^3 + x^4 + ... = x + x^3 + x^5 + ... + x^2 + x^4 + x^6 + ...$$

provided $|x| < 1$.

Suppose we have a bunch of groups, possibly infinitely many. By absolute convergence for any ε there is N such that, the tail after N terms is less than ε, and no matter how small ε is this holds if N is large enough. Also, any subset of the tail has size less than ε if you take the absolute value of all the terms, and thus that subset of the tail has absolute value less than ε in the original series by the triangle inequality which works for infinite sums. Let M be large enough such that the first N terms are in the first N terms of the first N groups. Then it follows that if we take the sum of the first M terms of the first M groups, this is within ε of the original sum, so as M goes to infinity this approaches the original sum.

□

**Fact.** It is easy to see that absolute convergence implies convergence: If for each $\varepsilon$ there is an M such that $\sum_{n=M+1}^{\infty} |a_n| < \varepsilon$ then by the triangle inequality we see that in fact $|\sum_{n=M+1}^{\infty} a_n| \leq \sum_{n=M+1}^{\infty} |a_n| < \varepsilon$.

*Remark.* With the counterexample series above, it fails because $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + ...$ (which is the sum of the absolute values of that series) tends to infinity. There are some nice proofs of this which we will see later, and are honestly way easier than the proof that we know it is true because the above proposition fails.

**Definition.** Suppose we have a power series like $a_0 + a_1 x + a_2 x^2 + a_3 x^3 + ...$, then its radius of convergence R is the largest value of $x$ such that $|x| < R$ implies that the power series converges absolutely. We know that $R = 1$ in the case of the "binomial theorem" taylor series from the ratio between terms idea, and we want to show that for any $|x| < R$ the actual derivative of the power series matches the term-by-term derivative.

**Lemma.** Let a and b be complex numbers and let n be a positive integer greater than 1, then

$$b^n - a^n - n(b-a)a^{n-1} = (b-a)^2(b^{n-2} + 2ab^{n-3} + 3a^2 b^{n-4} + ... + (n-1)a^{n-2}) \tag{3}$$

*Proof.* If $b = a$ this is obvious as $0 = 0$. Otherwise, note that

$$\frac{b^n - a^n}{b - a} = b^{n-1} + ab^{n-2} + a^2 b^{n-3} + ... + a^{n-1}$$

We can differentiate both sides with respect to a to get

$$\frac{(b-a)(-na^{n-1}) + b^n - a^n}{(b-a)^2} = b^{n-2} + 2ab^{n-3} + 3a^2 b^{n-4} + ... + (n-1)a^{n-2}$$

Multiplying both sides by $(b-a)^2$ gives the desired result.

□

**Lemma.** Let $a_0 + a_1 z + a_2 z^2 + a_3 z^3 + ...$ be a power series with radius of convergence R and suppose that $|z| < R$. Then the thing that we will are trying to prove is the derivative (ie, the series $\sum n a_n z^{n-1}$) converges absolutely.

*Proof.* Pick r such that $|z| < r < R$. Then since $r < R$, $\sum |a_n| r^n$ converges. Therefore, let C be an upper bound for the size of the terms in $\sum |a_n| r^n$. Now,

$\sum n|a_n z^{n-1}| = \sum n|a_n| r^{n-1} \left(\frac{|z|}{r}\right)^{n-1} \leq \frac{C}{r} \sum n \left(\frac{|z|}{r}\right)^{n-1}$. However, it is clear that the ratio between consecutive terms will be $\frac{n}{n-1}\frac{|z|}{r} \to \frac{|z|}{r} < 1$, so the series will converge. Hence $\sum n|a_n z^{n-1}|$ converges so the result follows.

□

Note that by applying the result above again and dividing by 2, $\sum \binom{n}{2} a_n z^{n-2}$ converges absolutely.

**Definition.** Let $f$ and $h$ be functions of $x$ and suppose that $h$ is not 0 if $x$ is close to 0, then we say that $f$ is $o(h)$ as $x \to 0$ (lower case o, the capital O means something else) if as $x \to 0$, $\frac{f}{h} \to 0$. Intuitively, this means $f$ is much smaller than $h$ for small values of $x$.

**Theorem.** The derivative of a power series equals the term by term derivative inside the radius of convergence.

*Proof.* Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \text{ and } g(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

We want to show that $f'(z) = g(z)$.

Since we want $\frac{f(z+h)-f(z)}{h} - g(z) \to 0$, we want $f(z+h) - f(z) - hg(z)$ to be $o(h)$. Note that this equal to

$$\sum_{n=2}^{\infty} a_n((z+h)^n - z^n - hnz^{n-1})$$

Since it is easy to see that the $n = 0$ and $n = 1$ terms would go to 0 (the $n = 1$ term would go to $(z+h) - z - h$)

Using equation (3) from earlier with $b = z + h, a = z$ gives that we want to consider

$$h^2 \sum_{n=2}^{\infty} a_n((z+h)^{n-2} + 2z(z+h)^{n-3} + ... + (n-1)z^{n-2})$$

We want the infinite series to be bounded so that we can conclude it is $o(h)$. To do this, pick r such that $|z| < r < R$ and h small enough so that $|z + h| \leq r$. The absolute value of the last infinite series is then bounded by

$$\sum_{n=2}^{\infty} |a_n|(r^{n-2} + 2r^{n-2} + ... + (n-1)r^{n-2}) = \sum_{n=2}^{\infty} |a_n| \begin{pmatrix} n \\ 2 \end{pmatrix} (r^{n-2})$$

But this is bounded.

□

*Remark.* Differentiation of power series multiplies the ratio between consecutive terms by $\frac{n}{n-1}$ and shifts the terms by 1, so if the ratio approached something less than 1 or more than 1 that property will still hold after differentiation, therefore the radius of convergence of the differentiated or antidifferentiated power series is the same as the radius of convergence of the original power series (This argument works for the "usual" functions where it is the case that the ratio between consecutive terms approaches something, but the statement is true more generally even in some cases where this argument does not work and the terms do not approach a ratio, but you do not need to know this for A level).

# 20 (Appendix C) The Lebesgue integral

The above definition of the integral (The Riemann integral) is perfectly good for continuous functions – we just showed that. However the modern definition of the integral (The Lebesgue integral) is something which is easy to see (once we actually define it) is the same as the Riemann integral for continuous functions, and even functions that are defined by many continuous parts, but it is defined for more diverse functions and we will need this definition when we justify stuff in statistics in level 6. We need some preliminary definitions.

**Definition.** The supremum is the least upper bound of some set. The infimum is the greatest lower bound of some sets (if the sets are bounded in the relevant direction).

**Definition.** Suppose I have a sequence of functions $f_1(x), f_2(x), f_3(x), ...$ each bounded above by something, then the pointwise supremum (if it exists) is the function $g(x)$ such that for each a in the domain of $f$, $g(a)$ is the supremum of the set $\{f_1(a), f_2(a), f_3(a), ...\}$

**Definition.** An indicator function of a subset of the real numbers is a function that takes the value 1 if the number is in the set and 0 otherwise. For example, suppose I have a set that consists of the numbers between 0 and 1 and the numbers between 1.5 and 2, then figure 12 shows what the graph of its indicator function would look like.

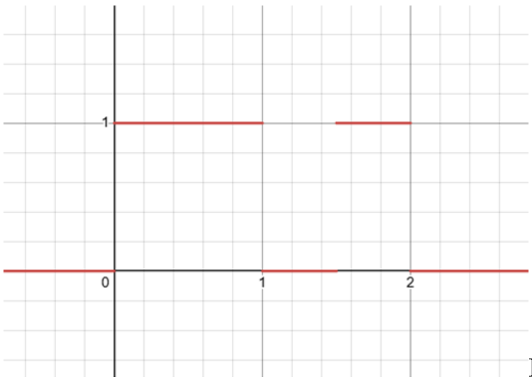

Figure 12

Now I will try to underestimate the integral $\int_0^3 (3x - x^2)dx$ by summing multiples of indicator functions. We define the integral of an indicator function as the length of the set in question (I will shortly discuss how I am defining length). Figure 13 will show how I am doing this approximation.

I could give a silly underestimate like 0, or I could use indicator functions of exotic sets like the cantor set. But here I will show a (relatively) simple way to underestimate it.

In figure 13 I am doing an example of underestimating the function by indicator functions. The exact formula for figure 13 is

$$\sum_{r=1}^{14} \text{Indicator}\left(1.5 - \frac{r}{10} < x < 1.5 + \frac{r}{10}\right)\left(0.31 - \frac{r}{50}\right)$$

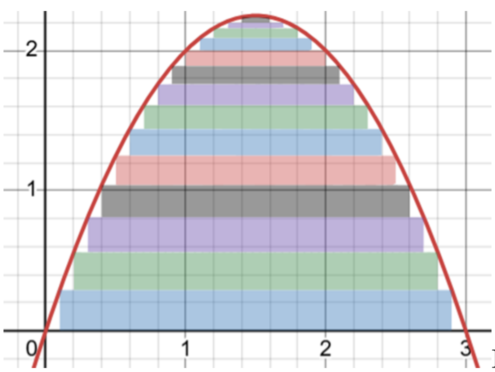

Figure 13

Note the contrast to the Riemann integral which approaches it by means of approximating with vertical rectangles, and always uses rectangles where this can use indicator functions of more exotic sets, like the set of all rational numbers between 0 and 1 (The "length" of this set is 0 but we will not prove this here).

The indicator function of the set of rationals between 0 and 1 is lebesgue integrable but not riemann integrable.

As for the definition of length, we will not define it fully precisely here. I believe that trying to give the rigorous definition would make everything here more complicated than it needs to be for the intended audience and wouldn't add much to your understanding of the proofs we will do that use this. It is also true that not every subset of the reals has a well defined length, however in practice when we use this the length of the sets we need will be obvious, and in level 6 when we work with probability distributions we will redefine the length as the probability of being within that set, which is a proper definition for all the sets we will work with even though there are some very weird sets where this fails, but luckily we will not do any theorems that claim something holds for any arbitrary set so this is not a problem.

Notice how the image above is an underestimate since our linear combination of indicator functions never exceeds our function. The integral is then defined as the supremum of ALL possible such underestimates (where the indicator functions have a well defined length, or are "measurable"). That's the definition of the Lebesgue integral.

**Definition.** A simple function is a sum of multiples of finitely many indicator functions.

The lebesgue integral is the supremum of the integrals of simple functions that underestimate the function we are trying to integrate.

**Theorem.** The Riemann integral on continuous functions gives the same value as the Lebesgue one

*Proof.* Each of the rectangles from the riemann integral can be considered to be a simple function and we just want the least upper bound of combinations of areas of those that are below the function. Adding in non-rectangle indicators to underestimate a Riemann integrable function does not increase the least upper bound and I will explain why by an example.

The example in figure 13 has a Riemann integral value of 4.5, so I can make a simple function out of rectangles of area 4.5000000000001 that is everywhere larger than that parabola and **cover** the parabola (by the Riemann integrability property of continuous functions) and say: Hey, if you want a larger integral than 4.5000000000001 you're gonna have to go outside this cover at some point even though you're trying to underestimate the function, so your least upper bound cannot exceed 4.5000000000001, and I can do that for values as close to 4.5 as I like, so the least upper bound of the area of simple functions is indeed 4.5 for the above example.

$\square$